Dynamical Genesis of Complex Structure on Graphs: Neimark–Sacker Bifurcation and Non-Abelian Holonomy

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Abstract

We study the discrete (inertial) Kuramoto model on finite graphs with signed couplings and explicit gauge frustration. We show that a supercritical *Neimark–Sacker bifurcation* generically creates a 2D invariant center manifold endowed with a canonical almost-complex structure $\mathcal{J}$. The rotation matrix J on the center manifold induces $\mathcal{J} := J/\omega$, defining a U(1) principal bundle over the graph. When frustrated triads carry nonzero curvature, a simple obstruction lemma motivates a non-abelian SU(2) lift, where discrete curvature is interpreted as the angle of an SU(2) rotation.

We derive explicit "spark conditions" for K_3 and for two-range K_4 motifs. We prove that the explicit-Euler discretization is essential for this genesis, as it introduces a dt^2 scaling term in the characteristic polynomial that enables unit-circle crossings. For K_4, tuning ring versus diagonal stiffness closes a spectral gap and can place two oscillatory instabilities near each other, creating a codimension-2 *availability* for resonant mixing on a 4D center manifold.

Numerically, we validate the NS mechanism on a directed, frustrated K_3. For frustrated K_4 parameter sets, we use a fixed-point–anchored audit (solve θ^*, compute stiffness spectrum σ, predict K^*, then verify by full Jacobian sweeps). This reveals a robust *multi-step NS ladder*. In inverse-designed near-degenerate regimes, nonlinear simulations often collapse to a single dominant frequency or transition to hyperchaos, indicating that the geometry exists only within a specific inertial window bounded by "spark" and "fracture."

1 Introduction and model

Let $G = (V, E)$ be a finite graph. We introduce a scalar coupling strength $K \in \mathbb{R}$ and signed edge couplings $\hat{\kappa}_{vw}$. To capture geometric frustration explicitly, we introduce an antisymmetric edge 1-cochain $\alpha_{vw} \in \mathbb{R}/2\pi\mathbb{Z}$. The discrete Kuramoto map with frustration is:

$$\theta_v(t+1) = \theta_v(t) + \Delta_v + \frac{K}{\deg(v)} \sum_{w \sim v} \hat{\kappa}_{vw} \sin\big(\theta_w(t) - \theta_v(t) - \alpha_{vw}\big) \pmod{2\pi}, \tag{1}$$

*With contributions by Grok, ChatGPT, Gemini, and Claude (listed as *contributors* in Zenodo metadata; sole human author of record).

where $\alpha_{vw} = -\alpha_{wv}$. The inertial extension evolves angles θ_v and velocities v_v via an explicit Euler update (dt, γ):

$$\theta_v(t+1) = \theta_v(t) + dt\, v_v(t), \tag{2}$$

$$v_v(t+1) = (1 - \gamma dt)\, v_v(t) + dt \cdot K\, \mathcal{F}_v(\theta), \tag{3}$$

where $\mathcal{F}_v$ is the *dimensionless* force term built from (1) without the overall factor K. **Convention:** For K_3 experiments we use degree-normalized sums ($\frac{1}{\deg(v)}$); for K_4 we use raw sums (1).

Discretization remark: when NS is (im)possible

- **Euler–Cromer (semi-implicit):** for the inertial linear mode, $|\lambda| = \sqrt{1 - \gamma\, dt}$, pinning complex modes strictly inside the unit circle; a Neimark–Sacker (NS) bifurcation is generically impossible.
- **Explicit Euler:** NS crossings become possible and are controlled by the fixed-point stiffness spectrum; see Proposition 5.5 and Corollary 5.3.

2 Neimark–Sacker bifurcation and emergent complex structure

Theorem 2.1 (Emergent complex structure)**.** *Consider the discrete inertial map (2)–(3) near a phase-locked fixed point. If the Jacobian has a simple pair of complex-conjugate eigenvalues $\lambda_\pm(K)$ crossing the unit circle with angle $\omega \in (0, \pi)$, a Neimark–Sacker bifurcation occurs. In the supercritical case, the 2D center manifold carries a canonical almost-complex structure $\mathcal{J} := J/\omega$ satisfying $\mathcal{J}^2 = -I$.*

3 Local-to-global U(1) bundle

We interpret the graph G as a base space for a discrete U(1) bundle with connection α_{ij}. Given phases θ, the **covariant edge difference** is $A_{ij}(t) := \theta_j(t) - \theta_i(t) - \alpha_{ij}$. The curvature on a triangle (i, j, k) is the holonomy $F_{ijk} := (\alpha_{ij} + \alpha_{jk} + \alpha_{ki}) \bmod 2\pi$.

4 Frustration and non-Abelian holonomy

Lemma 4.1 (Frustration obstruction)**.** *If $F_\triangle \not\equiv 0 \pmod{2\pi}$, the connection α is not exact and has nontrivial holonomy.*

Definition 4.2 (Consistent non-Abelian lift)**.** *A consistent lift is an $\mathfrak{su}(2)$-valued connection $\mathbf{A}_{ij}$ such that $\frac{1}{2}\,\mathrm{Tr}\,\mathbf{H}(\partial\triangle) = \cos(F_\triangle/2)$.*

Definition 4.3 (Trace gap diagnostic)**.** *Given a lifted triangle holonomy $\mathbf{H}(\partial\triangle) \in \mathrm{SU}(2)$ and curvature $F_\triangle$, define the* trace gap

$$\Delta_f(\triangle) \;:=\; \frac{1}{2}\,\mathrm{Tr}\;\mathbf{H}(\partial\triangle) \;-\; \cos(F_\triangle/2).$$

A perfectly aligned consistent lift yields $\Delta_f = 0$. Misalignment and non-commuting components can produce $|\Delta_f| = \mathcal{O}(1)$.

Remark 4.4 (BCH obstruction and axis misalignment). In practical diagnostics, the lift is expressed in a chosen frame. If the dynamically selected rotation axis is misaligned with this diagnostic frame, Baker–Campbell–Hausdorff commutators generically perturb the effective holonomy trace, producing $|\Delta_f| = \mathcal{O}(1)$ even when the underlying U(1) curvature is fixed. We report representative aligned vs. misaligned trace gaps in Table 1.

Conjecture 4.5. *For every frustrated triad, there exists a consistent lift where $\mathbf{H} = \exp(i\frac{F_\triangle}{2}\mathbf{n}\cdot\boldsymbol{\sigma})$.*

5 Mathematical derivation: the genesis of quaternionic structure

5.1 The inertial spark in the symmetric frustrated triangle

Proposition 5.1 (The genesis spark in K_3 (symmetric backbone))**.** *Consider $N = 3$ inertial oscillators with explicit Euler discretization (dt, γ) and degree-normalized couplings. Assume a symmetric locked state with equal covariant lags $\delta^* = (2\pi - \Phi)/3$. The critical coupling for the NS bifurcation is:*

$$K^* = \frac{2\gamma}{3dt \cos\left(\frac{2\pi-\Phi}{3}\right)}.$$

Proof. **Step 1 (Stiffness):** With degree normalization ($\deg = 2$), the tangent stiffness matrix M has off-diagonal entries $\frac{K}{2}\cos(\delta^*)$ and diagonal entries $-K\cos(\delta^*)$. **Step 2 (Spectrum):** This Laplacian has eigenvalues 0 and $-1.5K\cos(\delta^*)$ (multiplicity 2). Thus the structural eigenvalues are $\mu_{1,2} = -1.5K\cos(\delta^*)$. **Step 3 (Spark):** The NS condition for explicit Euler requires $\mu = -\gamma/dt$. Equating $-1.5K^*\cos(\delta^*) = -\gamma/dt$ yields $K^* = \frac{2\gamma}{3dt\cos(\delta^*)}$. □

Remark 5.2 (Experimental reality). Our numerical K_3 experiment uses detuning and directed asymmetry, making it a generic perturbation of this symmetric backbone. The observed $K_c \approx 1.93$ differs from the symmetric K^* prediction, but the NS mechanism remains robust.

Corollary 5.3 (The NS angle identity)**.** *At the exact moment of the spark ($|\lambda| = 1$) in explicit Euler inertial dynamics, the rotation angle ω is pinned by the inertial parameters:*

$$\cos\omega = 1 - \frac{\gamma dt}{2}.$$

Proof. For a single stiffness mode, the inertial Jacobian block has trace $2 - \gamma dt$. At the NS crossing $|\lambda| = 1$ and $\lambda \neq \pm 1$, $\lambda + \bar{\lambda} = 2\cos\omega = \mathrm{Tr}$, hence $\cos\omega = 1 - \gamma dt/2$. □

5.2 Multi-spark ladder and gap closing in two-range K_4

Proposition 5.4 (Spectral gap closing)**.** *Consider a two-range K_4 motif with **effective ring stiffness** $A = a\cos\delta_r$ and **effective diagonal stiffness** $B = b\cos\delta_d$. The alternating and cross eigenmodes become degenerate when $A = B$.*

Proof. The tangent stiffness matrix (without degree normalization) is:

$$M = K\begin{pmatrix} -(2A+B) & A & B & A \\ A & -(2A+B) & A & B \\ B & A & -(2A+B) & A \\ A & B & A & -(2A+B) \end{pmatrix}.$$

Eigenvalues are $\mu_{\text{alt}} = -4KA$ and $\mu_{\text{cross}} = -2K(A+B)$. Equating them gives $A = B$. □

Proposition 5.5 (Fixed-point NS ladder from stiffness spectrum)**.** *Let θ^* be a phase-locked equilibrium of the inertial explicit-Euler system (2)–(3), and let $S := D_\theta \mathcal{F}(\theta^*)$ be the Jacobian of the* dimensionless *force at θ^* with eigenvalues σ_j. For every* real negative *stiffness eigenvalue $\sigma_j < 0$, the corresponding inertial mode admits an NS crossing at*

$$K_j^* \;=\; \frac{\gamma}{dt\,(-\sigma_j)}.$$

Moreover, the NS angle at each such crossing is $\omega = \arccos(1 - \gamma dt/2)$ (Corollary 5.3).

Proof. Linearizing (2)–(3) on an eigenvector of S with eigenvalue σ gives the 2D update

$$\begin{pmatrix} \delta\theta_{t+1} \\ \delta v_{t+1} \end{pmatrix} = \begin{pmatrix} 1 & dt \\ dt\,K\sigma & 1-\gamma dt \end{pmatrix} \begin{pmatrix} \delta\theta_t \\ \delta v_t \end{pmatrix}.$$

The characteristic polynomial is $\lambda^2 - (2-\gamma dt)\lambda + (1-\gamma dt - dt^2 K\sigma) = 0$. At an NS crossing $|\lambda| = 1$ and $\lambda \neq \pm 1$, the determinant must equal 1, hence $1-\gamma dt - dt^2 K\sigma = 1 \Rightarrow K\sigma = -\gamma/dt$. If $\sigma < 0$, this yields $K^* = \gamma/(dt(-\sigma))$. □

Remark 5.6 (Notation: stiffness vs. structural eigenvalues). Some sections use the full (dimensional) force Jacobian $M := K\,S$. Its eigenvalues μ_j satisfy $\mu_j = K\,\sigma_j$. In particular, at an NS crossing, $\mu_j(K_j^*) = -\gamma/dt$.

5.3 The inertial ceiling: fracture and gain

The emergent geometry is empirically bounded by two critical thresholds: a lower "spark" where a complex pair first reaches $|\lambda| = 1$, and an upper "fracture" where the nonlinear dynamics transition into hyperchaos.

Proposition 5.7 (The genesis window (heuristic))**.** *Stable genesis of oscillatory structure requires that the discrete inertial update does not overshoot the effective constraint manifold selected by frustration and flux. Empirically, we observe a window*

$$K_{\text{spark}} < K < K_{\text{fracture}},$$

where K_{spark} is predicted by Proposition 5.5 and K_{fracture} is detected by the onset of persistent positive Lyapunov exponents and large trace gaps $|\Delta_f| = \mathcal{O}(1)$ (Definition 4.3).

6 Numerical confirmation (2025 update)

6.1 Protocol: fixed-point–anchored NS auditing

For K_4, reliable NS thresholds require computing the *actual equilibrium* θ^*, then its stiffness spectrum $\sigma = \text{spec}(D_\theta \mathcal{F}(\theta^*))$, and only then predicting K^* via Proposition 5.5. We validate by sweeping the full inertial Jacobian eigenvalues and tracking complex pairs.

Remark 6.1 (Detuning changes the equilibrium equation). If intrinsic frequencies $\Omega \neq 0$ are included, a locked state satisfies $\Omega + K\,\mathcal{F}(\theta^*) = 0$, not $\mathcal{F}(\theta^*) = 0$. Fixed-point solvers and stiffness audits must be performed on this modified equation.

6.2 K_3 directed frustrated test: NS and angle identity

We confirm the NS mechanism in a perturbed K_3 with directed asymmetry and detuning (explicit Euler inertia). The observed oscillation frequency at the NS onset matches the pinned NS angle identity (Corollary 5.3) to numerical precision (Table 1).

6.3 K_4 parameter set A (dt=0.09999): multi-step NS ladder (fixed-point verified)

Parameter set A (frustrated, weakly directed K_4 with two-range weights) is:

$$dt = 0.09999,\ \gamma = 0.09620,\ \varepsilon_{\text{asym}} \approx 0.0327,\ w_{\text{diag}} = 2.6220,\ \sigma_2 = -0.091384,\ \phi = 2.09480,$$

with raw diagonal weight $b_{\text{raw}} = w_{\text{diag}} + \sigma_2 \approx 2.530616$ and no degree normalization.

Using the fixed-point solver and stiffness audit pipeline, we obtain

$$\sigma \approx \{-6.65594564,\ -4.41310031,\ -1.75902540,\ 0\},$$

hence predicted crossings

$$K_1^* \approx 0.14454689, \quad K_2^* \approx 0.21800914, \quad K_3^* \approx 0.54694845.$$

Full Jacobian sweeps confirm these thresholds. In nonlinear runs in the post-second-crossing regime (e.g. $K \approx 0.2405$), Lyapunov diagnostics and spectra are consistent with strong mode locking: one near-zero (gauge-orthogonal) direction and the remaining exponents negative, indicating an attracting T^1 rather than a robust T^2.

Remark 6.2 (Relation to earlier Zenodo analysis (v2.0)). Zenodo version v2.0 (DOI 10.5281/zenodo.17735085) reported two near-zero Lyapunov exponents in a specific K_4 regime and interpreted this as possible T^2 dynamics. The present fixed-point–anchored audits and systematic Lyapunov scans indicate that this interpretation was premature: the long-time regime corresponds to T^1 locking (one gauge-orthogonal neutral exponent), while the second "near-zero" exponent is consistent with a finite-time transient.

6.4 K_4 parameter set B (dt=0.096): verified thresholds and Lyapunov scan

For the explicit-Euler inertial K_4 with parameters

$$dt = 0.096,\ \gamma = 0.102,\ \varepsilon_{\text{asym}} = 0.028,\ w_{\text{diag}} = 2.55,\ \sigma_2 = -0.1,\ \phi = 2.0948,$$

the fixed-point stiffness spectrum yields

$$\sigma \approx \{-6.47852557,\ -4.26558616,\ -1.78843429,\ 0\}$$

and hence the predicted crossings

$$K_1^* \approx 0.16400337, \quad K_2^* \approx 0.24908652, \quad K_3^* \approx 0.59409507.$$

Full Jacobian sweeps confirm these K^*.

To probe the long-time regime, we perform a Lyapunov scan over $K \in [0.2, 0.62]$ using gauge-orthogonal exponents (QR/Benettin method, $q = 6$). We observe:

- For $K \lesssim 0.3053$, a T^1 regime: one near-zero exponent (gauge-orthogonal) and all others negative.
- For $K \gtrsim 0.3053$, a hyperchaotic regime: two positive exponents persist across the scan range.

A bisection refinement brackets the transition near $K \in [0.305290, 0.305386]$.

6.5 Discussion: what controls locking vs. hyperchaos?

We observe three qualitative outcomes in K_4: T^1 locking (one occupied frequency), hyperchaos (multiple positive Lyapunov exponents), and a lack of persistent T^2 despite codimension-2 availability. Empirically, the long-time regime appears sensitive to:

- **Stiffness gap** $|\sigma_1 - \sigma_2|$: smaller gaps bring crossings closer but also strengthen nonlinear mode competition.
- **Frequency commensurability:** near-rational ratios between dominant spectral lines promote mode locking and suppression of a second incommensurate frequency.

- **Cross-saturation in the normal form:** strong cubic cross-coupling can slave one mode to the other even when both are linearly unstable.
- **Inertial gain and "fracture":** beyond a narrow window, the system transitions directly from locking to hyperchaos, consistent with the genesis window in Proposition 5.7.

A robust T^2 would plausibly require both Diophantine (sufficiently irrational) frequency ratios *and* weak cross-coupling, which may be non-generic for these frustrated, strongly coupled motifs.

6.6 Validation summary

Table 1: Validation summary (fixed-point–anchored numerics).

System	Quantity	Predicted	Observed	Status
K_3	NS angle $\omega_{\mathrm{id}} = \arccos(1 - \gamma dt/2)$	0.17342 rad	0.1734 rad	✓
K_3	K_c (symmetric backbone)	backbone only	1.93 (perturbed)	—
K_4 (set A)	K_1^* (from σ)	0.1445469	sweep-confirmed	✓
K_4 (set A)	K_2^* (from σ)	0.2180091	sweep-confirmed	✓
K_4 (set A)	K_3^* (from σ)	0.5469485	sweep-confirmed	✓
K_4 (set A)	Attractor type (tested post-K_2^*)	locking allowed	T^1 (single neutral)	✓
K_4 (set B)	K_1^* (from σ)	0.1640034	sweep-confirmed	✓
K_4 (set B)	K_2^* (from σ)	0.2490865	sweep-confirmed	✓
K_4 (set B)	K_3^* (from σ)	0.5940951	sweep-confirmed	✓
K_4 (set B)	Chaos onset (Lyap scan)	—	$K \in [0.305290, 0.305386]$	✓
K_4 (set B)	Long-time regime for $K \gtrsim 0.3053$	—	hyperchaos (≥ 2 LEs > 0)	✓
Lift diagnostic	Trace gap (aligned lift) Δ_f	0	$\sim 10^{-16}$	✓
Lift diagnostic	Trace gap (misaligned lift) Δ_f	$\mathcal{O}(1)$	≈ 0.5	✓

7 Conjectures and conclusion

Conjecture 7.1 (Dimensional ladder — refined)**.** *For $k \in \{1, 3, 7\}$, there exists a minimal frustrated graph G_k such that:*

1. ***Spectral prerequisite:*** *the inertial dynamics admit a codimension-$\lceil k/2 \rceil$ NS environment (availability of $\lceil k/2 \rceil$ complex pairs crossing $|\lambda| = 1$).*
2. ***Algebraic availability:*** *at resonance, the center-manifold commutant admits a nontrivial division-algebra-like mixing structure (e.g. $\mathfrak{su}(2)$-type mixing in the $k = 3$ case).*
3. ***Occupied attractor dimension:*** *the observed attractor may have dimension $m \leq \lceil k/2 \rceil$; strong mode locking can reduce the occupied dynamics (often to T^1), while inertial fracture can lead directly to hyperchaos.*

Status: $k = 1$ is realized on K_3 (T^1). For $k = 3$, K_4 supports codimension-2 NS availability, but in our tested parameter sets the long-time dynamics are typically locked (T^1) or hyperchaotic rather than robustly T^2.

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